

Corporate Finance (in lingua inglese) - A. Y. 2025/2026 - Topic 5c: NPV and other investment criteria - solution

TEXT and SOLUTION (through the use of Excel)

William Ltd is planning to build a new plant. The expenditure for the new equipment will be € 2.000. The new plant will substitute an old one. This old plant, completely depreciated and with no salvage value, will no longer produce goods for the next 3 years and thus no longer generate revenues and costs of, respectively, 300 and 50 a year. The new equipment will be depreciated over a period of 4 years, using straight-line depreciation. Yearly revenues and expenses (not including depreciation) concerning the new plant will be 1.000 and 200, respectively, from year 1 throughout the entire project life. The workers will use, as their accommodation for the entire period starting in year 1, a building owned by the firm that is rented and the rent is € 35 a year. The firm forecasts, for the new plant, an annual value of the working capital of 10% of its annual revenues in each year, from year 1 to 3, while the working capital will be zero in the fourth year. No working capital is required for the old plant over the period analysed. The salvage value of the new plant at the end of the fourth year will be € 150. The cost of capital for the new project is 8%, the tax rate is 30% and its life is 4 years. Decide on the basis of the NPV rule if the firm should build the new plant.

		Period				
Year		0	1	2	3	4
Panel A Capital Investment						
1	Cash flow from investment in fixed assets	- 2.000,00				
2	Sales of fixed assets					150,00
3	Less tax on sales					45,00
4	Cash flow from capital investment (1+2+3)	- 2.000,00				105,00

Panel B Operating Cash Flow						
5	Revenues		665,00	665,00	665,00	965,00
6	Expenses*		150,00	150,00	150,00	200,00
7	Depreciation		500,00	500,00	500,00	500,00
8	Pretax profit (5 - 6 - 7)		15,00	15,00	15,00	265,00
9	Tax at 30%		4,50	4,50	4,50	79,50
10	Profit after tax (8 - 9)		10,50	10,50	10,50	185,50
11	Operating cash flow (10 + 7)		510,50	510,50	510,50	685,50

Panel C Investment in Working Capital						
12	Working capital		100,00	100,00	100,00	0,00
13	Change in working capital		100,00	0,00	0,00	-100,00
14	Cash flow from investment in w. c . (-13)		-100,00	0,00	0,00	100,00

Panel D Project Valuation						
15	Total project cash flow (4 + 11 + 14)	- 2.000,00	410,50	510,50	510,50	890,50
16	Discount factor at 8%	1,00	0,93	0,86	0,79	0,74
17	Discounted cash flows (15 x 16)	- 2.000,00	380,09	437,67	405,25	654,54
18	NPV (sum of discounted cash flows)	- 122,44				

* Not including depreciation

Alternative method for calculating the discounted cash flows (by putting all the data in a single formula)

Calculations of the discounted cash flows at 8%					
Period	0	1	2	3	4
Formula	$-2000,00/(1+8\%)^0$	$410,50/(1+8\%)^1$	$510,50/(1+8\%)^2$	$510,50/(1+8\%)^3$	$890,50/(1+8\%)^4$
Discounted cash flows	-2000,00	380,09	437,67	405,25	654,54

The firm should not build the new plant as the NPV is lesser than 0.